testbook

GET IT ON Google Play

BEL FTE

Previous Year Paper (Electrical) 18 Jan, 2025 Shift 1

testbook

testbook

GET IT ON Google Play

QUESTION PAPER
Electrical [SHIFT - 1]

Exam Date : 18/01/2025

Time : 10:00 AM - 12:00 PM

Part I - Aptitude

Question 1

Question Id : 1

Directions: *Study the passage below to answer the question that follows.*

Passage
There are some men who seem to be always on the lookout for trouble and, to tell the truth, they are seldom disappointed. Listening to such men, one would think that this world is one of the stormiest and most disagreeable places. Yet, after all, it is not such a bad place, and the difficulty is often in the man who is too thin-skinned. On the other hand, the man who goes out expecting people to be like himself, kind and brotherly, will be surprised at the kindness he meets even in the most unlikely quarters. A smile is apt to be met with a responsive smile while the sneer is just as apt to provoke a snarl. Men living in the same neighbourhood may live vastly different lives. But it is not the neighbourhood which is quarrelsome, but the man within us. And we have it in our power to change our neighbourhood into a pleasant one by simply changing our own ways.

The passage is about

Answer :

	Option Id
(A) our disagreeable and hostile world.	1001
(B) a kind and pleasant world.	1002
(C) our indifferent and unresponsive world.	1003
(D) the world and what one makes of it.	1004

Right Answer :
the world and what one makes of it.

Right Option Id : 1004

Question 2

Question Id : 2

Directions: *Study the passage below to answer the question that follows.*

Passage
There are some men who seem to be always on the lookout for trouble and, to tell the truth, they are seldom disappointed. Listening to such men, one would think that this world is one of the stormiest and most disagreeable places. Yet, after all, it is not such a bad place, and the difficulty is often in the man who is too thin-skinned. On the other hand, the man who goes out expecting people to be like himself, kind and brotherly, will be surprised at the kindness he meets even in the most unlikely quarters. A smile is apt to be met with a responsive smile while the sneer is just as apt to provoke a snarl. Men living in the same neighbourhood may live vastly different lives. But it is not the neighbourhood which is quarrelsome, but the man within us. And we have it in our power to change our neighbourhood into a pleasant one by simply changing our own ways.

"..... they are seldom disappointed." The statement denotes that such men

Answer :

	Option Id
(A) welcome difficulties as a morale booster.	2001
(B) do not have to face any trouble.	2002
(C) manage to keep unruffled in the face of discomforts.	2003
(D) generally, do not fail to come across troubles.	2004

Right Answer :
generally do not fail to come across troubles.

Right Option Id : 2004

Question 3

Question Id : 3

Directions: *Study the passage below to answer the question that follows.*

Passage
There are some men who seem to be always on the lookout for trouble and, to tell the truth, they are seldom disappointed. Listening to such men, one would think that this world is one of the stormiest and most disagreeable places. Yet, after all, it is not such a bad place, and the difficulty is often in the man who is too thin-skinned. On the other hand, the man who goes out expecting people to be like himself, kind and brotherly, will be surprised at the kindness he meets even in the most unlikely quarters. A smile is apt to be met with a responsive smile while the sneer is just as apt to provoke a snarl. Men living in the same neighbourhood may live vastly different lives. But it is not the neighbourhood which is quarrelsome, but the man within us. And we have it in our power to change our neighbourhood into a pleasant one by simply changing our own ways.

The author's own view of the world is that it is

testbook

GET IT ON Google Play

Answer :

Option Id

(A) one of the loveliest and quietest places. 3001

(B) an unpleasant and turbulent place. 3002

(C) one's own excessive sensitivity that makes it a bad place. 3003

(D) a sordid place for those who suffer in life. 3004

Right Answer :

one's own excessive sensitivity that makes it a bad place.

Right Option Id : 3003

Question 4

Question Id : 4

Directions: *Study the passage below to answer the question that follows.*

Passage

There are some men who seem to be always on the lookout for trouble and, to tell the truth, they are seldom disappointed. Listening to such men, one would think that this world is one of the stormiest and most disagreeable places. Yet, after all, it is not such a bad place, and the difficulty is often in the man who is too thin-skinned. On the other hand, the man who goes out expecting people to be like himself, kind and brotherly, will be surprised at the kindness he meets even in the most unlikely quarters. A smile is apt to be met with a responsive smile while the sneer is just as apt to provoke a snarl. Men living in the same neighbourhood may live vastly different lives. But it is not the neighbourhood which is quarrelsome, but the man within us. And we have it in our power to change our neighbourhood into a pleasant one by simply changing our own ways.

Which of the following is opposite in meaning to the expression 'thin-skinned' as used in the passage?

Answer :

Option Id

(A) Insensitive 4001

(B) Intelligent 4002

(C) Awkward 4003

(D) Obstinate 4004

Right Answer :

Insensitive

Right Option Id : 4001

Question 5

Question Id : 5

Directions: *Study the passage below to answer the question that follows.*

Passage

There are some men who seem to be always on the lookout for trouble and, to tell the truth, they are seldom disappointed. Listening to such men, one would think that this world is one of the stormiest and most disagreeable places. Yet, after all, it is not such a bad place, and the difficulty is often in the man who is too thin-skinned. On the other hand, the man who goes out expecting people to be like himself, kind and brotherly, will be surprised at the kindness he meets even in the most unlikely quarters. A smile is apt to be met with a responsive smile while the sneer is just as apt to provoke a snarl. Men living in the same neighbourhood may live vastly different lives. But it is not the neighbourhood which is quarrelsome, but the man within us. And we have it in our power to change our neighbourhood into a pleasant one by simply changing our own ways.

"On the other hand, ... unlikely quarters." The statement shows that people's reaction to our attitude is

Answer :

Option Id

(A) generally indifferent. 5001

(B) surprisingly synchronous. 5002

(C) often adverse. 5003

(D) mainly favourable. 5004

Right Answer :

surprisingly synchronous.

Right Option Id : 5002

Question 6

Question Id : 6

If the sum of two numbers is 22 and the sum of their squares is 404, then product of these two numbers is _____.

Answer :

Option Id

(A) 40 6001

(B) 44 6002

(C) 80 6003

(D) 88 6004

testbook
GET IT ON Google Play

Right Answer :
40

Right Option Id : 6001

Question 7

Question Id : 7

A trader invested a certain sum of money in a bank for 6 years at 7 per cent per annum simple interest. He found that after 6 years, the interest amounted to Rs.812 is less than the sum lent. How much money he invested?

Answer :

Option Id

(A) Rs.1050 — 7001
(B) Rs.1400 — 7002
(C) Rs.1125 — 7003
(D) Rs.1350 — 7004

Right Answer :
Rs.1400

Right Option Id : 7002

Question 8

Question Id : 8

The difference of compound interest on Rs.1600 for 1 year at 20% per annum when compounded half yearly and quarterly is _____.

Answer :

Option Id

(A) Rs.10.50 — 8001
(B) Rs.13.42 — 8002
(C) Rs.11.61 — 8003
(D) Rs.8.81 — 8004

Right Answer :
Rs.8.81

Right Option Id : 8004

Question 9

Question Id : 9

Two trains start from Jodhpur and Jaipur towards each other at 9 a.m. with their respective speeds of 70 km. and 85 km. an hour. If the two trains cross each other at 12 noon, the distance between two stations is _____.

Answer :

Option Id

(A) 315 km — 9001
(B) 385 km — 9002
(C) 420 km — 9003
(D) 465 km — 9004

Right Answer :
465 km

Right Option Id : 9004

Question 10

Question Id : 10

A can do a piece of work in 12 days. B is 60% more efficient than A. The number of days it takes B to do the same piece of work is _____.

Answer :

Option Id

(A) 6 days — 10001
(B) $6\ ^1/_4$ days — 10002
(C) $7\ ^1/_2$ days — 10003
(D) 8 days — 10004

Right Answer :
$7\ ^1/_2$ days

Right Option Id : 10003

Question 11

Question Id : 11

Sheetal is the daughter of Amit. Juli and Amit are a married couple. Juli has two children, Abhay and a daughter. Kunal is the husband of Amit's only daughter. How is Abhay related to Kunal?

Answer :

Option Id

(A) Brother-in-law — 11001
(B) Brother — 11002
(C) Sister-in-law — 11003
(D) Can't be determined — 11004

Right Answer :
Brother-in-law

Right Option Id : 11001

Question 12

Question Id : 12

Seven students are standing in a column in the school assembly. Disha is standing three places ahead of Parul. Arun is standing in the middle. Mohit and Lakshman stand somewhere ahead of Arun. Kavita stands right after Nirmal, and Lakshman stands right after Disha. Who is standing between Parul and Kavita?

Answer :

Option Id

(A) Disha — 12001
(B) Nirmal — 12002
(C) Mohit — 12003
(D) Arun — 12004

Right Answer :
Nirmal

Right Option Id : 12002

Question 13

Question Id : 13

In a certain code, CONTRACT is written as TNOCTCAR. How is ATTRACTION written in that code?

Answer :

Option Id

(A) ARTATNOITC — 13001
(B) ARTTANOITC — 13002
(C) ARTTAONITC — 13003
(D) ARTTANOCIT — 13004

Right Answer :
ARTTANOITC

Right Option Id : 13002

Question 14

Question Id : 14

Amit is 6 years older than his brother Babu who is twice as old as his nephew Chirag. If the total of the ages of the three is 66 years, then how old is Babu?

Answer :

Option Id

(A) 15 — 14001
(B) 14 — 14002
(C) 12 — 14003
(D) 24 — 14004

Right Answer :
24

Right Option Id : 14004

Question 15

Question Id : 15

One orange and three apples cost Rs.180 but one apple and three oranges cost Rs.300. What is the cost of one apple?

Answer :

Option Id

(A) Rs.90 — 15001
(B) Rs.60 — 15002
(C) Rs.30 — 15003

(D) None of these

15004

Right Answer :
Rs.30

Right Option Id : 15003

Question 16

Question Id : 16

Dry ice is prepared with _____.

Answer :

Option Id

(A) Carbon dioxide — 16001
(B) Nitrogen — 16002
(C) Oxygen — 16003
(D) None of these — 16004

Right Answer :
Carbon dioxide

Right Option Id : 16001

Question 17

Question Id : 17

Who built the rock temple of Kailasa at Ellora caves in Maharashtra?

Answer :

Option Id

(A) Pallavas — 17001
(B) Cholas — 17002
(C) Rashtrakutas — 17003
(D) Palas — 17004

Right Answer :
Rashtrakutas

Right Option Id : 17003

Question 18

Question Id : 18

Who among the following started the Arya Samaj?

Answer :

Option Id

(A) Gopal Krishna Gokhale — 18001
(B) Raja Ram Mohan Roy — 18002
(C) Swami Vivekananda — 18003
(D) Swami Dayanand Saraswati — 18004

Right Answer :
Swami Dayanand Saraswati

Right Option Id : 18004

Question 19

Question Id : 19

The Chief Justice and other Judges of a High Court in India are appointed by the

Answer :

Option Id

(A) President of India. — 19001
(B) Chief Justice of India. — 19002
(C) Governor of the state concerned. — 19003
(D) Chief Minister of the state concerned. — 19004

Right Answer :
President of India.

Right Option Id : 19001

Question 20

Question Id : 20

Which of the following celestial bodies bears, 'The Sea of Tranquility' and 'The Ocean of Storms'?

Answer :

Option Id

(A) Mars — 20001
(B) Venus — 20002
(C) Moon — 20003
(D) Sun — 20004

Right Answer :
Moon

Right Option Id : 20003

Question 21

Question Id : 21

Directions: *Study the following information and answer the question that follows.*

At a flood relief centre, poor families are handed over dry rations by donors. The dry ration consists of different items viz. Rice bags, Atta bags, Pulses packets, Cooking oil pouches and Biscuit packets. Total number of all 5 items distributed in April 2024 first week was 1650. 24% of all items were Rice bags, $1/6^{th}$ were Atta bags, 14% were Pulses packets. Remaining items distributed are either Cooking oil pouches or Biscuit packets. Number of Biscuit packets distributed were 50 more than Cooking oil pouches distributed.

What is the difference between the total number of Cooking oil pouches and Rice bags distributed and the number of Pulses packets distributed?

Answer :

Option Id

(A) 534 — 21001
(B) 524 — 21002
(C) 511 — 21003
(D) 514 — 21004

Right Answer :
514

Right Option Id : 21004

Question 22

Question Id : 22

Directions: *Study the following information and answer the question that follows.*

At a flood relief centre, poor families are handed over dry rations by donors. The dry ration consists of different items viz. Rice bags, Atta bags, Pulses packets, Cooking oil pouches and Biscuit packets. Total number of all 5 items distributed in April 2024 first week was 1650. 24% of all items were Rice bags, $1/6^{th}$ were Atta bags, 14% were Pulses packets. Remaining items distributed are either Cooking oil pouches or Biscuit packets. Number of Biscuit packets distributed were 50 more than Cooking oil pouches distributed.

Number of Cooking oil pouches distributed is approximately what percent of the total number of Pulses packets and Biscuits packets put together?

Answer :

Option Id

(A) 63 — 22001
(B) 55 — 22002
(C) 59 — 22003
(D) 51 — 22004

Right Answer :
55

Right Option Id : 22002

Question 23

Question Id : 23

Directions: *Study the following information and answer the question that follows.*

At a flood relief centre, poor families are handed over dry rations by donors. The dry ration consists of different items viz. Rice bags, Atta bags, Pulses packets, Cooking oil pouches and Biscuit packets. Total number of all 5 items distributed in April 2024 first week was 1650. 24% of all items were Rice bags, $1/6^{th}$ were Atta bags, 14% were Pulses packets. Remaining items distributed are either Cooking oil pouches or Biscuit packets. Number of Biscuit packets distributed were 50 more than Cooking oil pouches distributed.

What is the total number of Atta Bags, Pulses packets and Biscuits packets distributed?

Answer :

Option Id

(A) 907 — 23001
(B) 917 — 23002

(C) 925
(D) 905

23003
23004

Right Answer :
905

Right Option Id : 23004

Question 24

Question Id : 24

Directions: *Study the following information and answer the question that follows.*

At a flood relief centre, poor families are handed over dry rations by donors. The dry ration consists of different items viz. Rice bags, Atta bags, Pulses packets, Cooking oil pouches and Biscuit packets. Total number of all 5 items distributed in April 2024 first week was 1650. 24% of all items were Rice bags, $1/6^{th}$ were Atta bags, 14% were Pulses packets. Remaining items distributed are either Cooking oil pouches or Biscuit packets. Number of Biscuit packets distributed were 50 more than Cooking oil pouches distributed.

What is the respective ratio between the number of Biscuit packets and the number of Pulses packets distributed by the flood relief centre?

Answer :
(A) 17:11
(B) 11:17
(C) 19:11
(D) 19:13

Option Id
24001
24002
24003
24004

Right Answer :
19:11

Right Option Id : 24003

Question 25

Question Id : 25

Directions: *Study the following information and answer the question that follows.*

At a flood relief centre, poor families are handed over dry rations by donors. The dry ration consists of different items viz. Rice bags, Atta bags, Pulses packets, Cooking oil pouches and Biscuit packets. Total number of all 5 items distributed in April 2024 first week was 1650. 24% of all items were Rice bags, $1/6^{th}$ were Atta bags, 14% were Pulses packets. Remaining items distributed are either Cooking oil pouches or Biscuit packets. Number of Biscuit packets distributed were 50 more than Cooking oil pouches distributed.

If 24% of the Atta bags are of poor quality, what is the number of the same which are of Good quality to be distributed?

Answer :
(A) 209
(B) 215
(C) 219
(D) 225

Option Id
25001
25002
25003
25004

Right Answer :
209

Right Option Id : 25001

Part II - Technical

Question 26

Question Id : 26

In the node voltage method, which node is the reference node?

Answer :
(A) Node with maximum voltage
(B) Node with highest resistance
(C) Node with minimum current
(D) Node with most elements tied to it

Option Id
26001
26002
26003
26004

Right Answer :
Node with most elements tied to it

Right Option Id : 26004

Question 27

Question Id : 27

If the magnetic flux in a magnetic circuit increases when current and the number of turns increase, what is the quantity that directly determines this relationship?

Answer :

Option Id

(A) Magnetizing force — 27001
(B) Magnetic reluctance — 27002
(C) Magneto motive force — 27003
(D) Magnetic field intensity — 27004

Right Answer :
Magneto motive force

Right Option Id : 27003

Question 28

Question Id : 28

Which of the following conditions will result in a decrease in magnetic reluctance of a core material?

Answer :

Option Id

(A) Increasing the mean length of the magnetic circuit — 28001
(B) Decreasing the relative permeability of the material — 28002
(C) Increasing the cross-sectional area of the core — 28003
(D) Decreasing the magnetizing force applied — 28004

Right Answer :
Increasing the cross-sectional area of the core

Right Option Id : 28003

Question 29

Question Id : 29

What resolves the paradox of needing infinite sources and destinations in a circuit?

Answer :

Option Id

(A) Continuous pathway — 29001
(B) Infinite wire length — 29002
(C) Endless electrons — 29003
(D) Infinite power sources — 29004

Right Answer :
Continuous pathway

Right Option Id : 29001

Question 30

Question Id : 30

What is the primary function of the core in a transformer?

Answer :

Option Id

(A) Provide path for the magnetic flux — 30001
(B) Minimize eddy current losses — 30002
(C) Increase the efficiency of the windings — 30003
(D) Store electrical energy — 30004

Right Answer :
Provide path for the magnetic flux

Right Option Id : 30001

Question 31

Question Id : 31

Where are the carbon brushes housed in a DC generator?

Answer :

Option Id

(A) Circular holder — 31001
(B) Rectangular holder — 31002
(C) Armature core — 31003
(D) Field coil — 31004

testbook
GET IT ON Google Play

Right Answer :
Rectangular holder

Right Option Id : 31002

Question 32

Question Id : 32

What does the yoke in a DC generator provide a channel for?

Answer :

Option Id

(A) Electric current — 32001
(B) Magnetic flux — 32002
(C) Mechanical energy — 32003
(D) Heat — 32004

Right Answer :
Magnetic flux

Right Option Id : 32002

Question 33

Question Id : 33

What does the line integral of the magnetic field intensity (H) about any closed path equal according to Ampere's Law?

Answer :

Option Id

(A) Magnetic flux enclosed by the path — 33001
(B) Electric field induced in the loop — 33002
(C) Current enclosed by the path — 33003
(D) The sum of electric charges inside the loop — 33004

Right Answer :
Current enclosed by the path

Right Option Id : 33003

Question 34

Question Id : 34

What is the role of binary representation in digital electronics?

Answer :

Option Id

(A) It is used to represent sound and image data. — 34001
(B) It helps design various circuits of a computer system. — 34002
(C) It is used to control external devices. — 34003
(D) It converts analog signals to digital. — 34004

Right Answer :
It helps design various circuits of a computer system.

Right Option Id : 34002

Question 35

Question Id : 35

What happens when an alternating current flows through the primary coil of a transformer?

Answer :

Option Id

(A) A constant magnetic field is created in the core. — 35001
(B) A varying magnetic flux is induced in the core. — 35002
(C) The current is directly passed to the secondary coil. — 35003
(D) The primary coil generates a direct current (DC) in the secondary coil. — 35004

Right Answer :
A varying magnetic flux is induced in the core.

Right Option Id : 35002

Question 36

Question Id : 36

What happens when a conductor moves parallel to the magnetic field as per the First Law of Electromagnetic Induction?

Answer :

Option Id

(A) An emf is induced in the conductor. 36001
(B) The galvanometer needle shows deflection. 36002
(C) The magnetic flux through the conductor changes. 36003
(D) No emf is induced in the conductor. 36004

Right Answer :
No emf is induced in the conductor.

Right Option Id : 36004

Question 37

Question Id : 37

You are required to design a sequential system in digital electronics. Which requirement is a must to define it?

Answer :

Option Id

(A) Continuous signals 37001
(B) No memory element 37002
(C) Past inputs 37003
(D) Logic gates 37004

Right Answer :
Past inputs

Right Option Id : 37003

Question 38

Question Id : 38

You are analyzing the architecture of an RISC processor as part of a project. You need to determine how many clock cycles it takes for the processor to execute a single instruction. How many clock cycles does it take to complete this task?

Answer :

Option Id

(A) 5 clock cycles 38001
(B) 1 clock cycle 38002
(C) 2 clock cycles 38003
(D) 0 clock cycle 38004

Right Answer :
1 clock cycle

Right Option Id : 38002

Question 39

Question Id : 39

What is required for a machine with field coils to generate a magnetic field?

Answer :

Option Id

(A) A permanent magnet 39001
(B) A constant current 39002
(C) A current to flow in the coils 39003
(D) A rotating rotor 39004

Right Answer :
A current to flow in the coils

Right Option Id : 39003

Question 40

Question Id : 40

What does the Maximum Power Transfer Theorem state when applied to DC networks?

Answer :

Option Id

(A) Load resistance is half the network resistance. 40001
(B) Load resistance equals network resistance. 40002
(C) Load resistance is greater than network resistance. 40003
(D) Load resistance is less than network resistance. 40004

Right Answer :
Load resistance equals network resistance.

Right Option Id : 40002

Question 41

Question Id : 41

You are working on a distributed computing system where you need to understand the role of various components. One component, the transputer, is used in parallel processing systems. What is its primary function in a computing system?

Answer :	Option Id
(A) Store data	41001
(B) Manage power distribution	41002
(C) Perform processing with local memory	41003
(D) Control input/output devices	41004

Right Answer :
Perform processing with local memory

Right Option Id : 41003

Question 42

Question Id : 42

How does an ADC function to convert analog signals into a digital format for processing?

Answer :	Option Id
(A) By performing Fourier analysis on the signal	42001
(B) By sampling voltage at fixed intervals	42002
(C) By converting the signal to a frequency	42003
(D) By only quantizing the signal into discrete steps	42004

Right Answer :
By sampling voltage at fixed intervals

Right Option Id : 42002

Question 43

Question Id : 43

You are working on the design of a microprocessor-based system and need to understand the role of the interrupt controller. What does it do in a microprocessor?

Answer :	Option Id
(A) Process main program through interrupts	43001
(B) Handle interrupt requests	43002
(C) Prioritize tasks based on execution time	43003
(D) Process multiple instructions simultaneously	43004

Right Answer :
Handle interrupt requests

Right Option Id : 43002

Question 44

Question Id : 44

What determines the magnitude of the self-induced emf in a coil?

Answer :	Option Id
(A) The steady-state current in the coil	44001
(B) The instantaneous value of the flux linkage	44002
(C) The rate of change of current in the coil	44003
(D) The coil's resistance	44004

Right Answer :
The rate of change of current in the coil

Right Option Id : 44003

Question 45

Question Id : 45

How is the magnetic field generated in a self-excited field generator?

Answer :

	Option Id
(A) By using external DC power	45001
(B) By the armature voltage supplied to the field winding	45002
(C) By a separate battery connected to the field winding	45003
(D) By stator induction from the armature winding	45004

Right Answer :

By the armature voltage supplied to the field winding

Right Option Id : 45002

Question 46

Question Id : 46

You are working on a project that involves designing an electric motor. A conductor will need to experience a force as part of the motor's operation. What are the two necessary conditions to produce a force on a conductor?

Answer :

	Option Id
(A) Carrying current and within a magnetic field	46001
(B) High voltage and vacuum	46002
(C) Insulated conductor and electric field	46003
(D) High resistance and magnetic field	46004

Right Answer :

Carrying current and within a magnetic field

Right Option Id : 46001

Question 47

Question Id : 47

Why must the field winding conductor be sized properly in a series motor?

Answer :

	Option Id
(A) To reduce current loss	47001
(B) To carry the load current consumption	47002
(C) To increase the voltage supply	47003
(D) To control the speed	47004

Right Answer :

To carry the load current consumption

Right Option Id : 47002

Question 48

Question Id : 48

What happens to the self-induced emf in a coil when steady-state DC current flows in the circuit?

Answer :

	Option Id
(A) It opposes the applied voltage indefinitely	48001
(B) It aligns with the applied voltage	48002
(C) It vanishes	48003
(D) It amplifies the current flow	48004

Right Answer :

It vanishes

Right Option Id : 48003

Question 49

Question Id : 49

What happens when the load resistance of a shunt generator is reduced beyond a certain point?

Answer :

	Option Id
(A) The load current continues to increase	49001
(B) The terminal voltage increases due to increased current	49002
(C) The load current reduces due to armature reaction and IR drop	49003

(D) The external characteristics of the generator remain unchanged

49004

Right Answer :

The load current reduces due to armature reaction and IR drop

Right Option Id : 49003

Question 50

Question Id : 50

What is the condition of voltage in a series generator at no-load?

Answer :

Option Id

(A) High voltage — 50001

(B) Small voltage — 50002

(C) Unstable voltage — 50003

(D) Constant voltage — 50004

Right Answer :

Small voltage

Right Option Id : 50002

Question 51

Question Id : 51

In wind turbine technology, what principle explains the extraction of energy from wind?

Answer :

Option Id

(A) Bernoulli's principle — 51001

(B) Betts' Law — 51002

(C) Newton's third law — 51003

(D) Joule's Law — 51004

Right Answer :

Betts' Law

Right Option Id : 51002

Question 52

Question Id : 52

Which material is known for its potential to significantly increase solar cell efficiency beyond the silicon limit?

Answer :

Option Id

(A) Gallium arsenide (GaAs) — 52001

(B) Copper indium gallium selenide (CIGS) — 52002

(C) Graphene — 52003

(D) Perovskite — 52004

Right Answer :

Perovskite

Right Option Id : 52004

Question 53

Question Id : 53

What is the primary role of the small signal in DC chopper circuits for controlling power electronic devices?

Answer :

Option Id

(A) To increase power output — 53001

(B) To switch power components on or off — 53002

(C) To convert AC to DC — 53003

(D) To store energy — 53004

Right Answer :

To switch power components on or off

Right Option Id : 53002

Question 54

Question Id : 54

Which of the following best describes the behavior of non-linear components, where the current is not directly proportional to the applied voltage?

Answer :

Option Id

(A) The voltage increases exponentially with current — 54001
(B) Resistance is constant over a wide range of voltage — 54002
(C) Resistance varies with voltage and current — 54003
(D) Current is inversely proportional to resistance — 54004

Right Answer :

Resistance varies with voltage and current

Right Option Id : 54003

Question 55

Question Id : 55

If a solar panel has an efficiency of 15% and receives 600 W/ m^2 of solar power, what is the output power per square meter?

Answer :

Option Id

(A) 60 W/ m^2 — 55001
(B) 30 W/ m^2 — 55002
(C) 120 W/ m^2 — 55003
(D) 90 W/ m^2 — 55004

Right Answer :

90 W/ m^2

Right Option Id : 55004

Question 56

Question Id : 56

What is the main advantage of using HVDC transmission over AC transmission for long distances?

Answer :

Option Id

(A) Lower construction costs — 56001
(B) Simpler switchgear — 56002
(C) Less electromagnetic interference — 56003
(D) Higher transmission efficiency — 56004

Right Answer :

Higher transmission efficiency

Right Option Id : 56004

Question 57

Question Id : 57

In a power system, what is the typical impact of increasing the transmission line length on the system stability?

Answer :

Option Id

(A) Increases both transient and steady-state stability — 57001
(B) Increases transient stability but decreases steady-state stability — 57002
(C) Decreases transient stability but increases steady-state stability — 57003
(D) Decreases both transient and steady-state stability — 57004

Right Answer :

Decreases both transient and steady-state stability

Right Option Id : 57004

Question 58

Question Id : 58

A wind turbine operates in an area with an average wind speed of 7 m/s. Using the air density of 1.225 kg/m^3, what is the wind power density (W/m^2) at this speed?

Answer :

Option Id

(A) 400 W/m^2 — 58001
(B) 150 W/m^2 — 58002
(C) 300 W/m^2 — 58003
(D) 200 W/m^2 — 58004

testbook
GET IT ON Google Play

Right Answer :
200 W/m^2

Right Option Id : 58004

Question 59

Question Id : 59

You are working on designing a robotic arm for an industrial automation system. The design requires a motor that can provide reliable performance, especially in terms of low maintenance and high efficiency, as the robotic arm will be used continuously for long periods of time. Given the design constraints and requirements of the robotic arm, what is the significant advantage of using permanent magnet motors in this application?

Answer : Option Id

(A) High power factor 59001
(B) Low maintenance 59002
(C) High starting torque 59003
(D) Low operating temperature 59004

Right Answer :
Low maintenance

Right Option Id : 59002

Question 60

Question Id : 60

You are studying the behavior of magnetic fields inside a solenoid as part of a physics lab experiment. The solenoid in question is long, with a uniform winding and a constant current running through it. In magnetostatics, how does the magnetic field behave inside it?

Answer : Option Id

(A) It is zero 60001
(B) It varies linearly with the length of the solenoid 60002
(C) It decreases exponentially with the radius of the solenoid 60003
(D) It is uniform and parallel to the axis of the solenoid 60004

Right Answer :
It is uniform and parallel to the axis of the solenoid

Right Option Id : 60004

Question 61

Question Id : 61

What is the purpose of impedance matching in RF (Radio Frequency) circuits?

Answer : Option Id

(A) To reduce power consumption 61001
(B) To minimize power transfer 61002
(C) To increase voltage 61003
(D) To minimize signal reflection 61004

Right Answer :
To minimize signal reflection

Right Option Id : 61004

Question 62

Question Id : 62

What is the significance of the Crest Factor in waveform analysis?

Answer : Option Id

(A) It indicates the symmetry of the waveform 62001
(B) It measures the harmonic content 62002
(C) It determines the power factor 62003
(D) It is the ratio of peak value to RMS value 62004

Right Answer :
It is the ratio of peak value to RMS value

Right Option Id : 62004

Question 63

Question Id : 63

When viewed from its output terminals, what is the equivalent of a two-terminal active network with a voltage source and resistance?

Answer :

Option Id

(A) Constant voltage & series resistance — 63001
(B) Constant current & parallel resistance — 63002
(C) Constant current & series resistance — 63003
(D) Voltage source & parallel resistance — 63004

Right Answer :

Constant voltage & series resistance

Right Option Id : 63001

Question 64

Question Id : 64

Why are higher supply frequencies, such as 400 Hz or 1000 Hz, used in airborne equipment?

Answer :

Option Id

(A) To reduce the weight of electromagnetic devices for the same power output. — 64001
(B) To increase the power output of the electromagnetic devices. — 64002
(C) To reduce the hysteresis losses in magnetic materials. — 64003
(D) To improve the efficiency of alternating magnetization. — 64004

Right Answer :

To reduce the weight of electromagnetic devices for the same power output.

Right Option Id : 64001

Question 65

Question Id : 65

Why is a four-wire measurement method used in precision resistance measurements?

Answer :

Option Id

(A) To reduce the effects of inductance — 65001
(B) To increase the measurement speed — 65002
(C) To measure high voltages — 65003
(D) To eliminate the effect of wire resistance — 65004

Right Answer :

To eliminate the effect of wire resistance

Right Option Id : 65004

Question 66

Question Id : 66

For a type 0 control system with a position constant K_p of 199, what is the steady-state error for a step input of magnitude 10 units?

Answer :

Option Id

(A) 0.005 units — 66001
(B) 0.5 units — 66002
(C) 0.1 units — 66003
(D) 0.05 units — 66004

Right Answer :

0.05 units

Right Option Id : 66004

Question 67

Question Id : 67

Given a Bode plot with a phase cross-over frequency at 10 rad/s and the gain at this frequency is -20 dB, what is the gain margin?

Answer :

Option Id

(A) 0 dB — 67001
(B) 10 dB — 67002
(C) -20 dB — 67003

(D) 20 dB

67004

Right Answer :
20 dB

Right Option Id : 67004

Question 68

Question Id : 68

What is the primary reason for the phenomenon of magnetic hysteresis in a magnetic material?

Answer :

Option Id

(A) The magnetizing force leads to the magnetic flux density. 68001
(B) The magnetic flux density lags behind the magnetizing force. 68002
(C) The alternating current produces rotational magnetization. 68003
(D) The supply frequency is higher than 50 Hz. 68004

Right Answer :
The magnetic flux density lags behind the magnetizing force.

Right Option Id : 68002

Question 69

Question Id : 69

What happens when the two ends of a coil are connected across a DC voltage source in a magnetic field?

Answer :

Option Id

(A) The coil starts to heat up 69001
(B) A force is exerted on the coil, causing it to move 69002
(C) The coil vibrates without moving 69003
(D) No interaction occurs 69004

Right Answer :
A force is exerted on the coil causing it to move

Right Option Id : 69002

Question 70

Question Id : 70

Eddy currents in the armature core lead to power loss in a machine. What is its effect on the machine's efficiency?

Answer :

Option Id

(A) Reduces magnetic field and increases vibrations 70001
(B) Increases temperature and decreases efficiency 70002
(C) Causes mechanical vibrations 70003
(D) Enhances system efficiency 70004

Right Answer :
Increases temperature and decreases efficiency

Right Option Id : 70002

Question 71

Question Id : 71

What issue does the point starter address in a DC motor's field regulator?

Answer :

Option Id

(A) Motor disconnection 71001
(B) Overvoltage 71002
(C) Low voltage 71003
(D) No effect 71004

Right Answer :
Motor disconnection

Right Option Id : 71001

Question 72

Question Id : 72

In power electronics, what is the principle of operation of an interleaved boost converter?

Answer :

Option Id

(A) Alternating between different power paths to improve conversion efficiency — 72001
(B) Using a single high-capacity inductor to increase energy storage — 72002
(C) Sequentially operating multiple converters to step up the voltage — 72003
(D) Combining two or more boost converters in parallel to share the load and reduce ripple — 72004

Right Answer :

Combining two or more boost converters in parallel to share the load and reduce ripple

Right Option Id : 72004

Question 73

Question Id : 73

What advantage does a totem-pole PFC circuit have in power supply design?

Answer :

Option Id

(A) It reduces the physical size of the power supply — 73001
(B) It enhances the power factor to near unity with high efficiency — 73002
(C) It provides galvanic isolation from the mains supply — 73003
(D) It simplifies the circuit design, reducing the number of components needed — 73004

Right Answer :

It enhances the power factor to near unity with high efficiency

Right Option Id : 73002

Question 74

Question Id : 74

How are power electronic devices, like DC choppers, generally regulated?

Answer :

Option Id

(A) Through a high-power input signal — 74001
(B) With a small, low-power control signal — 74002
(C) Via manual operation by an operator — 74003
(D) Using external power supplies for regulation — 74004

Right Answer :

With a small low-power control signal

Right Option Id : 74002

Question 75

Question Id : 75

How is the constant current determined in a network using Norton's Theorem?

Answer :

Option Id

(A) By the open-circuit voltage of the network — 75001
(B) By the current that flows when the terminals are short-circuited — 75002
(C) By the network's resistance — 75003
(D) By the parallel resistance of the network — 75004

Right Answer :

By the current that flows when the terminals are short-circuited

Right Option Id : 75002

Question 76

Question Id : 76

In comparison to conventional technology, what does SoC eliminate the need for?

Answer :

Option Id

(A) Complex software — 76001
(B) A circuit board — 76002

(C) Additional components
(D) Power supply

76003
76004

Right Answer :
Additional components

Right Option Id : 76003

Question 77

Question Id : 77

What is the primary reason that the B-H curve of non-magnetic materials, such as air, is a straight line?

Answer :

Option Id

(A) The permeability of non-magnetic materials is constant. 77001
(B) Non-magnetic materials saturate at low flux density. 77002
(C) The magnetizing force in non-magnetic materials depends on flux density. 77003
(D) Non-magnetic materials have a variable relative permeability. 77004

Right Answer :
The permeability of non-magnetic materials is constant.

Right Option Id : 77001

Question 78

Question Id : 78

Why does the B-H curve of magnetic materials, such as cast steel, show a nonlinear shape?

Answer :

Option Id

(A) Magnetic materials have a constant relative permeability. 78001
(B) The flux density in magnetic materials is independent of magnetizing force. 78002
(C) The relative permeability of magnetic materials varies with flux density. 78003
(D) The magnetic length remains constant in magnetic materials. 78004

Right Answer :
The relative permeability of magnetic materials varies with flux density.

Right Option Id : 78003

Question 79

Question Id : 79

According to the energy band theory, what is the conductivity of an intrinsic semiconductor at ambient temperature?

Answer :

Option Id

(A) Zero 79001
(B) High 79002
(C) Moderate 79003
(D) Infinite 79004

Right Answer :
Zero

Right Option Id : 79001

Question 80

Question Id : 80

You are working with two types of semiconductors in a circuit. You notice that one semiconductor (n-type) conducts electricity more easily than the other. What could be the reason behind this?

Answer :

Option Id

(A) It has a higher number of holes that facilitate conduction. 80001
(B) It has more free electrons due to doping with Group V elements. 80002
(C) It has a lower electrical resistance due to its crystal structure. 80003
(D) It has fewer free electrons than the p-type semiconductor but more than a pure semiconductor. 80004

Right Answer :
It has more free electrons due to doping with Group V elements.

Right Option Id : 80002

GET IT ON Google Play

Question 81

Question Id : 81

What does the second operand typically represent in an instruction with two operands?

Answer :

Option Id

(A) The destination of the data — 81001

(B) A register to store the result — 81002

(C) The source — 81003

(D) The address of the destination — 81004

Right Answer :

The source

Right Option Id : 81003

Question 82

Question Id : 82

How can combinational circuits be categorized?

Answer :

Option Id

(A) Closed-loop systems — 82001

(B) Open-loop systems — 82002

(C) Sequential systems — 82003

(D) Feedback systems — 82004

Right Answer :

Open-loop systems

Right Option Id : 82002

Question 83

Question Id : 83

In an induction motor, what part is the same as in a synchronous machine?

Answer :

Option Id

(A) The rotor construction — 83001

(B) The power supply — 83002

(C) The physical stator — 83003

(D) The rotor speed — 83004

Right Answer :

The physical stator

Right Option Id : 83003

Question 84

Question Id : 84

You are working on designing a communication system to process audio signals. Your professor instructed you to use a low-pass filter (LPF) to help avoid a problem. What could that problem be?

Answer :

Option Id

(A) Amplification — 84001

(B) Signal distortion — 84002

(C) Aliasing of the message signal — 84003

(D) Signal clipping — 84004

Right Answer :

Aliasing of the message signal

Right Option Id : 84003

Question 85

Question Id : 85

You are designing an asynchronous sequential circuit. As you work through the design, you need to select an appropriate memory element for your project. What would you select?

Answer :

Option Id

(A) Synchronous flip-flops — 85001

(B) Unclocked flip-flops — 85002

(C) Registers — 85003
(D) Latches controlled by clock pulses — 85004

Right Answer :
Unclocked flip-flops

Right Option Id : 85002

Question 86

Question Id : 86

You are designing a digital logic circuit and need to decide how many input values your AND gate should be able to accept. How many input values can it accept?

Answer : Option Id

(A) Only two inputs — 86001
(B) Two or more inputs — 86002
(C) Only one input — 86003
(D) At least three inputs — 86004

Right Answer :
Two or more inputs

Right Option Id : 86002

Question 87

Question Id : 87

What factors influence the fundamental frequency of a Quartz Crystal Oscillator, which is crucial for its stability and accuracy?

Answer : Option Id

(A) Voltage applied to the crystal — 87001
(B) Thickness and size of the quartz crystal — 87002
(C) Frequency of the input signal — 87003
(D) Magnetic field effects on the crystal — 87004

Right Answer :
Thickness and size of the quartz crystal

Right Option Id : 87002

Question 88

Question Id : 88

You are designing an open-loop control system. What will you directly supply to the plant or processing system to drive its operation?

Answer : Option Id

(A) Feedback signal — 88001
(B) Actuating signal — 88002
(C) Input signal — 88003
(D) Output signal — 88004

Right Answer :
Input signal

Right Option Id : 88003

Question 89

Question Id : 89

You are designing a multiplexer for a digital logic circuit in your project. How will you select its data inputs?

Answer : Option Id

(A) By the data inputs — 89001
(B) By the select lines — 89002
(C) By the output line — 89003
(D) By the control signal — 89004

Right Answer :
By the select lines

Right Option Id : 89002

Question 90

Question Id : 90

You are studying delta modulation for encoding analog signals into digital signals. What does the modulator create from the input signal in this?

Answer :

Option Id

(A) A constant signal — 90001
(B) A staircase-like signal — 90002
(C) A sine wave signal — 90003
(D) A square wave signal — 90004

Right Answer :

A staircase-like signal

Right Option Id : 90002

Question 91

Question Id : 91

You are studying electromagnetic waves and their different regions in the spectrum. Amidst which frequency range can infrared radiation be found in the spectrum?

Answer :

Option Id

(A) 300 Hz – 400 Hz — 91001
(B) 300 GHz – 400 THz — 91002
(C) 100 MHz – 1 GHz — 91003
(D) 1 THz – 10 THz — 91004

Right Answer :

300 GHz – 400 THz

Right Option Id : 91002

Question 92

Question Id : 92

You are performing an experiment on semiconductors to observe the effects of doping. When you introduce a pentavalent impurity to a pure semiconductor, what happens?

Answer :

Option Id

(A) Number of holes will increase — 92001
(B) Number of free electrons will increase — 92002
(C) Number of free electrons will decrease — 92003
(D) Forms a p-type semiconductor — 92004

Right Answer :

Number of free electrons will increase

Right Option Id : 92002

Question 93

Question Id : 93

You are designing a digital system that monitors two different voltage levels and needs to take action when one voltage exceeds the other. Which component would you use for this application?

Answer :

Option Id

(A) Comparator — 93001
(B) Operational amplifier — 93002
(C) Voltage regulator — 93003
(D) Digital-to-analog converter (DAC) — 93004

Right Answer :

Comparator

Right Option Id : 93001

Question 94

Question Id : 94

You are analyzing adaptive delta modulation to understand how it adjusts step-size for efficient signal representation. When is a larger step-size required for effective signal representation?

Answer :

Option Id

(A) When the modulating signal has a small slope — 94001

(B) When the modulating signal has a steep slope — 94002
(C) When the signal is constant — 94003
(D) When the signal is zero — 94004

Right Answer :
When the modulating signal has a steep slope

Right Option Id : 94002

Question 95

Question Id : 95

You are working on a project involving microwave communication. While studying the design of the system, you want to know how the distance covered by microwave signals relates to the characteristics of the antenna. How does the distance covered by microwave signals relate to the antenna?

Answer : Option Id

(A) Inversely proportional — 95001
(B) Independent — 95002
(C) Directly proportional — 95003
(D) Dependent on frequency — 95004

Right Answer :
Directly proportional

Right Option Id : 95003

Question 96

Question Id : 96

You are working on a data routing system where you need to manage multiple data streams efficiently. You are considering different components, and one of them is a multiplexer. What is its primary role in data routing applications?

Answer : Option Id

(A) To select and forward data from multiple input sources to a single output. — 96001
(B) To amplify signals for long-distance transmission. — 96002
(C) To store data in memory devices. — 96003
(D) To encrypt and secure data during transmission. — 96004

Right Answer :
To select and forward data from multiple input sources to a single output.

Right Option Id : 96001

Question 97

Question Id : 97

Which flip-flops are variations of the JK flip-flop?

Answer : Option Id

(A) SR and D flip-flops — 97001
(B) Master-slave D and T flip-flops — 97002
(C) RS and JK flip-flops — 97003
(D) SR and JK flip-flops — 97004

Right Answer :
Master-slave D and T flip-flops

Right Option Id : 97002

Question 98

Question Id : 98

You are learning about error detection techniques in digital communication. One of the methods you come across is parity, which helps ensure data integrity by checking the number of bits. Which of the following are the two types of parity?

Answer : Option Id

(A) Even and odd parity — 98001
(B) High and low parity — 98002
(C) Positive and negative parity — 98003
(D) Binary and decimal parity — 98004

Right Answer :
Even and odd parity

testbook
GET IT ON Google Play

Right Option Id : 98001

Question 99

Question Id : 99

You are designing a circuit where you need to isolate different stages of a signal while maintaining the integrity of the voltage. To achieve this, you decide to use a voltage buffer. What is the key feature of the output voltage in its circuit?

Answer :

	Option Id
(A) Output is lower than the input voltage	99001
(B) Output is equal to the input voltage	99002
(C) Output is higher than the input voltage	99003
(D) Output is variable depending on the input	99004

Right Answer :

Output is equal to the input voltage

Right Option Id : 99002

Question 100

Question Id : 100

You are working with an operational amplifier in a non-inverting amplifier configuration. As part of your design, you need to understand how the input voltage is applied to the op-amp in this setup. How is it applied to the given configuration to ensure proper amplification?

Answer :

	Option Id
(A) The input voltage is applied to the inverting pin.	100001
(B) The input voltage is applied to the non-inverting pin.	100002
(C) The input voltage is applied to both inverting and non-inverting pins equally.	100003
(D) The input voltage is applied to the feedback loop.	100004

Right Answer :

The input voltage is applied to the non-inverting pin.

Right Option Id : 100002

Question 101

Question Id : 101

You are designing a communication system to ensure the integrity of the data being transmitted. To detect errors in the transmission, you need to implement a method for generating a parity bit, but you're unsure where it should be applied in the system. Where should it be placed in the system, keeping the given requirements in mind?

Answer :

	Option Id
(A) At the receiver side to check for errors	101001
(B) At the transmitter side to generate and add a parity bit	101002
(C) In the processing unit for error correction	101003
(D) In the middle of the transmission to modify the data	101004

Right Answer :

At the transmitter side to generate and add a parity bit

Right Option Id : 101002

Question 102

Question Id : 102

You are designing an inverting summing amplifier with multiple input voltages. To handle the different input signals, you are considering adding resistors in parallel to the inverting input. What effect will this have on the output of the amplifier?

Answer :

	Option Id
(A) Least possible differences of input voltages come as output	102001
(B) Only max input voltage comes out as output	102002
(C) All input voltages are summed	102003
(D) The resistors act as filters for each input	102004

Right Answer :

All input voltages are summed

Right Option Id : 102003

Question 103

Question Id : 103

You are conducting an experiment in a laboratory where you are transitioning a material into its superconducting state. What phenomenon characterizes this transition?

Answer :

Option Id

(A) Infinite electrical resistance — 103001
(B) Zero electrical resistance — 103002
(C) Maximum thermal conductivity — 103003
(D) High magnetic permeability — 103004

Right Answer :
Zero electrical resistance

Right Option Id : 103002

Question 104

Question Id : 104

In the modified configuration of an inverting operational amplifier, what does the voltage divider on the non-inverting pin create?

Answer :

Option Id

(A) A unity gain amplifier — 104001
(B) A differential amplifier — 104002
(C) An integrator circuit — 104003
(D) A summing amplifier — 104004

Right Answer :
A differential amplifier

Right Option Id : 104002

Question 105

Question Id : 105

What tools are indispensable for designing very large-scale integrated circuits (VLSI)?

Answer :

Option Id

(A) Manual circuit testers and debugging tools — 105001
(B) EDA tools and design automation software — 105002
(C) Physical prototyping tools and fabrication equipment — 105003
(D) Simulation tools and verification software — 105004

Right Answer :
EDA tools and design automation software

Right Option Id : 105002

Question 106

Question Id : 106

How can a square wave be used to generate a triangle waveform?

Answer :

Option Id

(A) By applying a low-pass filter to the square wave — 106001
(B) By using a differentiating circuit with resistors and capacitors on the square wave — 106002
(C) By integrating the square wave signal with an operational amplifier — 106003
(D) By amplifying the square wave signal — 106004

Right Answer :
By integrating the square wave signal with an operational amplifier

Right Option Id : 106003

Question 107

Question Id : 107

You are working with a photodiode in a circuit designed to detect light. The photodiode generates current in response to light exposure, and you need to convert this current into a proportional output voltage for further processing in the system. Which component will you use for this task?

Answer :

Option Id

(A) Capacitor — 107001
(B) Resistor in the feedback loop — 107002
(C) Transistor — 107003
(D) Inductor — 107004

Right Answer :
Resistor in the feedback loop

Right Option Id : 107002

Question 108

Question Id : 108

You are analyzing a circuit that uses positive feedback, and you're trying to determine the type of resistance that this configuration effectively simulates. What type of resistance will be effectively simulated in this circuit?

Answer :

Option Id

(A) Positive resistance — 108001
(B) Negative resistance — 108002
(C) Dynamic resistance — 108003
(D) Infinite resistance — 108004

Right Answer :
Negative resistance

Right Option Id : 108002

Question 109

Question Id : 109

In memory addressing mode, what is typically required when operands are specified?

Answer :

Option Id

(A) Access to the data segment in main memory — 109001
(B) Use of a register to hold the operand — 109002
(C) Direct use of a CPU flag for operand selection — 109003
(D) Use of an immediate constant value — 109004

Right Answer :
Access to the data segment in main memory

Right Option Id : 109001

Question 110

Question Id : 110

What is the S-parameter S21 in a two-port network?

Answer :

Option Id

(A) Insertion loss — 110001
(B) Reflection coefficient — 110002
(C) Transmission coefficient — 110003
(D) Isolation — 110004

Right Answer :
Transmission coefficient

Right Option Id : 110003

Question 111

Question Id : 111

You are working with a sensor that is designed to measure a specific physical quantity, and you want to ensure that the sensor provides accurate readings across its entire range of measurements. The sensor has different output responses at various points within this range, and you need to choose a calibration technique that ensures accuracy across the full measurement range. Which calibration technique would be most useful in this case?

Answer :

Option Id

(A) Zero-point calibration — 111001
(B) Span calibration — 111002
(C) Multi-point calibration — 111003
(D) Environmental calibration — 111004

Right Answer :
Multi-point calibration

Right Option Id : 111003

Question 112

Question Id : 112

You are analyzing a parallel communication system where you've noticed that single-bit errors are more common in this system than in others. What could be the reason behind this?

Answer :

Option Id

(A) Sequential bit transfer allows for slower transmission. 112001

(B) Noise on individual lines increases the chance of a single-bit error. 112002

(C) Data encryption issues lead to errors in transmission. 112003

(D) Multiple channels interfere with the data transfer. 112004

Right Answer :

Noise on individual lines increases the chance of a single-bit error.

Right Option Id : 112002

Question 113

Question Id : 113

How is rotor voltage generated in an induction motor?

Answer :

Option Id

(A) Through a DC field current applied to the rotor 113001

(B) By physical connections in the rotor windings 113002

(C) Through electromagnetic induction in the rotor windings 113003

(D) By supplying external power directly to the rotor 113004

Right Answer :

Through electromagnetic induction in the rotor windings

Right Option Id : 113003

Question 114

Question Id : 114

How does a dual active bridge (DAB) converter operate?

Answer :

Option Id

(A) Using a single switch to modulate power flow 114001

(B) Through direct AC to AC conversion without isolation 114002

(C) By transferring energy through a high-frequency transformer linking two full-bridge converters 114003

(D) By converting AC input directly to DC output without modulation 114004

Right Answer :

By transferring energy through a high-frequency transformer linking two full-bridge converters

Right Option Id : 114003

Question 115

Question Id : 115

Which material is used to suspend the coil in a moving coil galvanometer, and why is it preferred?

Answer :

Option Id

(A) Aluminum, for its light weight and low torsional constant. 115001

(B) Phosphor-bronze, for its conductivity and low torsional constant. 115002

(C) Copper, for its excellent electrical conductivity. 115003

(D) Iron, for its magnetic properties. 115004

Right Answer :

Phosphor-bronze for its conductivity and low torsional constant.

Right Option Id : 115002

Question 116

Question Id : 116

What is the principle of operation of a superjunction MOSFET in power electronics?

Answer :

Option Id

(A) It uses a planar structure to increase the breakdown voltage. 116001

(B) It operates on the principle of charge cancellation to improve efficiency. 116002

(C) It integrates a Schottky barrier diode for faster switching. 116003

(D) It employs a vertical structure with alternating p and n regions to reduce on-resistance. 116004

Right Answer :
It employs a vertical structure with alternating p and n regions to reduce on-resistance.

Right Option Id : 116004

Question 117

Question Id : 117

You are analyzing the use of the longitudinal redundancy check (LRC) method in a communication system. The LRC is designed to detect certain types of errors during data transmission. What types of errors can be identified using this?

Answer : Option Id

(A) Two-bit errors 117001
(B) Only single-bit errors 117002
(C) Single-bit and burst errors 117003
(D) Errors occurring in specific columns 117004

Right Answer :
Single-bit and burst errors

Right Option Id : 117003

Question 118

Question Id : 118

What distinguishes a Class-D amplifier in power electronics?

Answer : Option Id

(A) Its ability to handle high frequencies only 118001
(B) Its high-efficiency operation by switching transistors on and off 118002
(C) Its direct conversion of AC to DC 118003
(D) Its use of linear voltage regulation 118004

Right Answer :
Its high-efficiency operation by switching transistors on and off

Right Option Id : 118002

Question 119

Question Id : 119

You are designing a power electronic circuit that involves controlling power loads. To optimize the circuit's performance, you need to select semiconductor devices with appropriate switching characteristics. What switching speed should your power semiconductor devices operate at as compared to signal devices?

Answer : Option Id

(A) Higher switching speeds 119001
(B) No switching operation 119002
(C) Identical switching speeds 119003
(D) Lower switching speeds 119004

Right Answer :
Lower switching speeds

Right Option Id : 119004

Question 120

Question Id : 120

In the nodal analysis method, if a circuit has "n" nodes, how many independent equations are required to solve the circuit with "n" nodes?

Answer : Option Id

(A) n 120001
(B) n+1 120002
(C) n−1 120003
(D) n×2 120004

Right Answer :
n−1

Right Option Id : 120003

Question 121

Question Id : 121

How does mesh analysis differ from nodal analysis in terms of the law used?

Answer :

Option Id

(A) Mesh analysis uses KCL, and nodal analysis uses KVL. 121001

(B) Nodal analysis uses KCL, and mesh analysis uses KVL. 121002

(C) Both mesh and nodal analysis use KCL. 121003

(D) Both mesh and nodal analysis use KVL. 121004

Right Answer :

Nodal analysis uses KCL and mesh analysis uses KVL.

Right Option Id : 121002

Question 122

Question Id : 122

What can damage power system equipment and connecting wires during electrical surges?

Answer :

Option Id

(A) Short circuits that disrupt the normal flow of current 122001

(B) High frequency noise that affects signal integrity 122002

(C) Excessive current flow that leads to overheating and damage 122003

(D) Undervoltage conditions that result in inadequate power supply 122004

Right Answer :

Excessive current flow that leads to overheating and damage

Right Option Id : 122003

Question 123

Question Id : 123

You are working on an electrical protection system for a large industrial facility. The system includes several components designed to prevent damage to critical equipment. One of these components is a protective relay. What role does it play in safeguarding the system?

Answer :

Option Id

(A) Controls voltage levels in the circuit 123001

(B) Detects fault conditions and triggers the circuit breaker 123002

(C) Regulates current flow during normal operation 123003

(D) Monitors temperature for overheating 123004

Right Answer :

Detects fault conditions and triggers the circuit breaker

Right Option Id : 123002

Question 124

Question Id : 124

How does an active power filter improve power quality in power electronics systems?

Answer :

Option Id

(A) By converting DC power to AC power efficiently 124001

(B) By stepping down the voltage to a safer level 124002

(C) By compensating for reactive power and reducing harmonics 124003

(D) By providing backup power during outages 124004

Right Answer :

By compensating for reactive power and reducing harmonics

Right Option Id : 124003

Question 125

Question Id : 125

What allows electrons in the valence band of a semiconductor to move to the conduction band?

Answer :

Option Id

(A) When there is a larger band gap than insulators 125001

(B) When external energy is provided 125002

(C) Due to the presence of impurities in the material 125003

(D) Due to the absence of charge carriers

125004

Right Answer :

When external energy is provided

Right Option Id : 125002